Sabah Afroz

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EDUCATION

North Carolina State University — Raleigh, NC ***Aug 2022 – Dec 2026***

Major: Computer Science, B.S

Relevant Coursework: Data Structures and Algorithms, Software Engineering, Neural Networks, Intro to Artificial Intelligence, Automated Learning and Data Analysis, Network Security, Game Design, Game Development

Affiliations: Women and Minority in Engineering Program, Women in Computer Science, TRIO Scholar, University Ambassador, Society of Sales Engineers

SKILLS

Languages: Java, Python, JavaScript, SQL

Frameworks & Tools: React, FastAPI, Spring Boot, Node.js, PHP, Laravel, HTML, CSS, REST APIs, MySQL, PostgreSQL, OpenSearch, AWS (Bedrock), Docker, Git, GitHub, Postman

AI/ML & Practices: pandas, Excel, LangChain/OpenAI API, AI Agent Workflows, MCP Integration, Prompt Engineering, OOP, JUnit, pytest, CI/CD, Agile, OWASP, Mermaid

EXPERIENCE

Web Developer — NC State Engineering IT, Raleigh, NC ***Aug 2026 – Present***

- Support development and maintenance of web applications and internal IT systems used across NC State Engineering.
- Work with PHP, Laravel/Laradock, MySQL, Docker, Git, Node.js, Yarn, and virtualized development environments.
- Apply OWASP-aligned secure coding practices, secure configuration, and environment hardening while contributing to website functionality and internal systems.
- Configure and evaluate AI models in OpenCode, including model access, token and context limits, usage settings, and research into model-specific capabilities and constraints.

R&D Software Intern — Baxter International Inc., Raleigh, NC ***June 2026 – Aug 2026***

- Developed Python-based AI agents that analyze software documentation, requirements, and source code to identify potential hazards, risks, and compliance concerns.
- Integrated AI-driven risk-detection workflows into CI/CD pipelines to support automated analysis throughout the software development lifecycle.
- Collaborated with software and R&D teams to research, prototype, and refine AI-assisted tools for medical-device software safety and compliance.
- Built an MCP-based context layer connecting ~26 repositories into PostgreSQL with incremental updates on pull requests/merges, avoiding full-repository reprocessing and improving retrieval scalability.

Technical Support Technician — Textile Creative & Technology Services, NC State University ***Jan 2026 – May 2026***

- Provide Tier 1–2 technical support for 1,000+ faculty, staff, and students across the Wilson College of Textiles.
- Troubleshoot hardware, software, classroom technology, device setup, software installation, and inventory issues.

Software Engineering Intern — UDPlatforms, Remote ***May 2025 – Aug 2025***

- Built responsive frontend features and reusable React components using JavaScript, HTML, and CSS.
- Connected frontend workflows to Python-based REST APIs, supporting user-profile functionality and application data flow.
- Troubleshot defects, tested API integrations, and collaborated with the engineering team through Agile development cycles.

PROJECTS

AI-Assisted Code Reviewer — Senior Design with Bandwidth, Team of 5 | Python, AWS ***Aug 2026 – Present***

- Building a GitHub pull request reviewer with three AI agents (analysis, review, verification) on AWS Bedrock that posts advisory comments.
- Designed a pipeline where code checks validate every AI finding before posting, so unsupported feedback never reaches developers.
- Implemented Python components with OpenSearch code retrieval; contributed to architecture, UML, and planning on a five-person team.

JobPing — Automated Job-Search Monitoring Platform | Python, REST APIs, Greenhouse, Lever

- Built a Python platform monitoring ~1,000 company career pages via Greenhouse/Lever, normalizing postings and filtering for entry-level/new-grad roles in target categories (software, AI, data, product).
- Built a scalable ingestion pipeline handling duplicate postings, inconsistent formats, and API failures across hundreds of sources.

NewRoots — Multilingual Resource Platform | AI Chatbot, Java, JavaScript

- Built an accessible resource platform and multilingual AI chatbot to help immigrants find healthcare, legal, food, and community services.

Casanova's Revenge — 3D Adventure Game | Godot, GDScript

- Developed inventory, crafting, item-interaction, UI, and day-night gameplay systems using reusable Godot scenes and GDScript.

LEADERSHIP

Engineering Career Fair Ambassador — North Carolina State University ***2023 – Present***

- Supported career fairs with 300+ companies; coordinated logistics, led volunteers, and resolved technical issues.

Undergraduate Researcher — North Carolina State University ***2023 – 2024***

- Researched VR applications for mental health; analyzed user-study data and presented findings at the Undergraduate Research Program.

Student Government — College of Engineering Senator ***2022 – 2023***

- Represented 10,000+ engineering students in policy discussions and campus initiatives; collaborated with committees to evaluate proposals and improve student services.